

Master Study Guide: Fundamentals of Thermodynamics and Heat Transfer (ME127)

Unit 1: Introduction and Thermodynamic Concepts - [5% - 7%] of Total Marks

A. Short Answer Questions (SAOs)

2082 Mid-Term Q1 (6 Marks):

Explain intensive, extensive and specific property with examples.

2081 End-Sem Q1 (8 Marks):

Explain intensive, extensive and specific property with examples. State and explain Zeroth law of thermodynamics.

2081 Mid-Term Q1 (2+6 Marks):

Define microscopic and macroscopic view point in thermodynamics. A cylinder with total volume of 2 m^3 has a movable piston as shown in figure below, when the piston is at one fourth of the length, both sides have the same specific volume of $8 \text{ m}^3/\text{kg}$. Determine the specific volumes of both sides when the piston is at middle of the cylinder. **[FIGURE PRESENT]**

- *Figure Description:* A horizontal cylinder of total volume 2 m^3 is divided into two chambers (say A on the left and B on the right) by a thin movable piston. Initially, the piston is at a distance $x = L/4$ from the left end, where L is the total length of the cylinder. Heat/piston movement is then allowed to bring the piston to the exact middle ($x = L/2$).

2080 End-Sem Q1 (4+2+2 Marks):

Define thermodynamic property. Explain intensive and extensive property with examples. Sketch polytropic processes with $n = 0, 1, \gamma$ and ∞ on a common P-V diagram. Derive an expression for work transfer for polytropic process. **[NOTE: Sketch & derivation are mapped to Unit 2 below.]**

2080 End-Sem Q2 (4+4 Marks):

What is quasi-static process? Differentiate between stored energy and transient energy with examples. **[NOTE: Stored vs transient energy is mapped to Unit 2 below.]**

2079 End-Sem Q1 (4+4 Marks):

Define macroscopic and microscopic viewpoint as applied to the study of thermodynamics. Also list their features. Differentiate between heat transfer and work transfer. Derive the mathematical expression for transfer for an isobaric process. **[NOTE: Heat vs work & isobaric work derivation are mapped to Unit 2 below.]**

2079 End-Sem Q2 (2+6 Marks):

Define thermodynamic state and thermodynamic process. A piston-cylinder device as shown in figure contains 0.05 m^3 of a gas initially at 200 kPa . At this state, a linear spring that has a spring constant of 150 kN/m is touching the piston but exerting no force on it. Now heat is transferred to the gas causing the piston to rise and to compress the spring until the volume inside the cylinder triples. If the cross-sectional area of the piston is 0.25 m^2 , determine the final pressure inside the cylinder. **[FIGURE PRESENT] [NOTE: The piston-cylinder spring program is mapped to Unit 2 below.]**

B. Long Answer Questions (LAQs)

No exclusive Long Answer Questions (15-markers) have been asked solely on Unit 1. Concepts are tested via the 8-mark questions in Group A.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Focuses entirely on foundational definitions and conceptual comparisons. Common comparisons include *macroscopic vs. microscopic viewpoints*, *intensive vs. extensive vs. specific properties*, and *quasi-static processes*.
- **Direct Repetitions:**
 - "Define intensive, extensive and specific property with examples" was asked in **2081 End-Sem Q1**, **2082 Mid-Term Q1**, and **2080 End-Sem Q1**.
 - "Define macroscopic and microscopic viewpoint" was asked in **2079 End-Sem Q1** and **2081 Mid-Term Q1**.
- **Key High-Priority Topics:**
 - State and explain Zeroth Law of Thermodynamics (defines temperature).
 - Difference between Macroscopic and Microscopic viewpoints.
 - Difference between Intensive, Extensive, and Specific properties (with typical civil engineering/materials examples like density, mass, volume).

Unit 2: Energy and Energy Transfer - [8% - 13%] of Total Marks

A. Short Answer Questions (SAQs)

2082 Mid-Term Q2 (6 Marks):

Define the terms saturation temperature, saturated liquid. In a non-flow process, a gas expands from volume 0.5 m^3 to a volume of 1 m^3 according to the relation $P = 3V^2 + 1/V$ where P is pressure in bar and V is volume in m^3 . Determine: (i) the pressure at the end of the expansion, and (ii) the work done by the gas in the expansion process in kJ. **[NOTE: Saturation definitions are mapped to Unit 3.]**

2081 End-Sem Q2 (4+4 Marks):

Sketch polytropic processes with $n = 0, 1, 1.4$ and ∞ on a common P-V diagram. Derive the mathematical expression for work transfer for an ideal gas undergoing polytropic process. In a non-flow process, a gas expands from volume 0.5 m^3 to a volume of 1 m^3 according to the law $P = 3V^2 + 1/V$ where P is pressure in bar and V is volume in m^3 . Determine: (i) The pressure at the end of the expansion and (ii) The work done by the gas in the expansion process, in kJ.

2081 Mid-Term Q2 (8 Marks):

Write expressions for displacement work transfer for constant pressure, constant temperature and polytropic process.

2081 Mid-Term Q7 (8 Marks):

Air (0.5 kg) in the piston cylinder device shown in Fig. has an initial pressure and temperature of 1 MPa and 500°C respectively. The system is cooled until the temperature reaches 50°C . It takes a pressure of 0.5 MPa to support the piston. Sketch the process on P-V and T-V diagrams and determine the total work transfer. **[Take $R = 287 \text{ J/kg K}$] [FIGURE PRESENT]**

- *Figure Description:* A vertically oriented piston-cylinder assembly containing "Air". The piston is free to move. There is an arrow pointing outward labeled "Q" representing heat rejection from the bottom of the cylinder.

2080 End-Sem Q1 (2+2 Marks - Part):

Sketch polytropic processes with $n = 0, 1, \gamma$ and ∞ on a common P-V diagram. Derive an expression for work transfer for polytropic process.

2080 End-Sem Q2 (4 Marks - Part):

Differentiate between stored energy and transient energy with examples.

2080 End-Sem Q4 (8 Marks):

A piston-cylinder device contains 0.05 m^3 of a gas initially at 200 kPa . At this state, a linear spring of 150 kN/m is touching the piston but exerting no force on it. Now heat is transferred to the gas, causing the piston to rise and to compress the spring until the volume inside the cylinder triples. If the cross-sectional area of the piston is 0.25 m^2 , determine: (i) The final pressure inside the cylinder, and (ii) The total work done by the gas.

2079 End-Sem Q1 (4 Marks - Part):

Differentiate between heat transfer and work transfer. Derive the mathematical expression for transfer for an isobaric process.

2079 End-Sem Q2 (6 Marks - Part):

A piston-cylinder device contains 0.05 m^3 of a gas initially at 200 kPa . At this state, a linear spring that has a spring constant of 150 kN/m is touching the piston but exerting no force on it. Now heat is transferred to the gas causing the piston to rise and to compress the spring until the volume inside the cylinder triples. If the cross-sectional area of the piston is 0.25 m^2 , determine the final pressure inside the cylinder. **[FIGURE PRESENT]**

- *Figure Description:* A vertical piston-cylinder assembly. Inside the cylinder is gas. A helical coil spring is attached to a rigid support at the top. The bottom of the spring is just touching the top face of the piston when the gas volume is 0.05 m^3 .

B. Long Answer Questions (LAQs)

2081 End-Sem Q7 (7+8 Marks - Part):

- **b.** Air undergoes three processes in series to form a cycle:
 - Process 1-2: constant volume cooling, $V_1 = V_2 = 2 \text{ m}^3$
 - Process 2-3: constant pressure heating, $P = 100 \text{ kPa}$, $V_3 = 10 \text{ m}^3$
 - Process 3-1: isothermal compression

Sketch the process on P-V diagram and determine the total work transfer.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *Highly numerical and graphical. It heavily tests displacement work ($\int P dV$) derivations and calculations across various thermodynamic processes.*
- **Direct Repetitions:**
 - *The non-flow expansion numerical: $P = 3V^2 + 1/V$ is **REPEATED: Identical values in 2081 End-Sem Q2 and 2082 Mid-Term Q2.***

- The Piston-Cylinder Spring problem (determining final pressure) is **REPEATED: Identical values in 2079 End-Sem Q2 and 2080 End-Sem Q4** (2080 also added a sub-question to find the total work done).
 - The sketch of polytropic processes ($n = 0, 1, 1.4/\gamma, \infty$) on a P - V diagram is **REPEATED in 2080 End-Sem Q1 and 2081 End-Sem Q2.**
 - **Key High-Priority Topics:**
 - Displacement work derivations for Isobaric, Isothermal, and Polytropic processes.
 - Spring-assisted piston cylinder problems (incorporating $F_{\text{spring}} = k \cdot x$ and $P_{\text{final}} = P_{\text{initial}} + k\Delta x/A$).
 - Properties of path functions (heat and work) vs. point functions.
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Unit 3: Properties of Pure Substance - [12% - 19%] of Total Marks

A. Short Answer Questions (SAOs)

2082 Mid-Term Q2 (Part):

Define the terms saturation temperature, saturated liquid.

2082 Mid-Term Q3 (6 Marks):

A vessel contains 2kg of saturated liquid water and saturated water vapor mixture at a temperature of 150°C. One third of the volume is saturated liquid and two third is saturated vapor. Determine the pressure, quality, volume, internal energy and enthalpy of the mixture.

2081 End-Sem Q3 (4+4 Marks):

Define the term moisture content, quality, superheated vapor and saturation pressure. A rigid vessel contains 8kg of water at 120°C. If 5kg of the water is in the liquid form and the rest in the vapor form. Determine: (a) The pressure in the vessel, (b) The volume of the tank, (c) The specific enthalpy.

2081 Mid-Term Q3 (4+4 Marks):

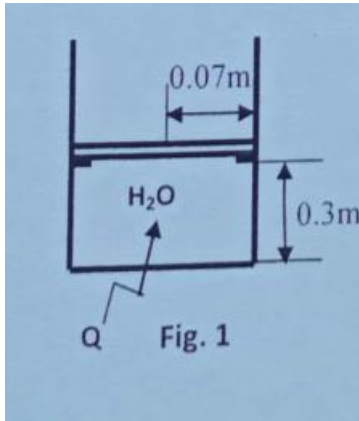
Derive an expression for specific volume of a two phase mixture in terms of quality. A 0.3 m³ rigid vessel contains 5Kg of water at 150kPa. Determine: (i) the temperature, (ii) the mass of each phase, (iii) the specific enthalpy.

2080 End-Sem Q3 (1+3+4 Marks):

Define pure substance. State and explain "state postulate". Derive the relation $v = v_f + x v_{fg}$ for two phase mixture.

2080 End-Sem Q5 (8 Marks):

A piston cylinder device shown in Fig.1 below contains water initially at a pressure of 125kPa with a quality of 50%. Heat is added to the system until it reaches a final temperature of 800°C. It takes a pressure of 600kPa to lift the piston from stops. Sketch the process on P-V and T-V diagram and determine: (i) The mass of H₂O in the system, and (ii) The total work transfer.



2079 End-Sem Q3 (2+3+3 Marks):

Define pure substance. Explain why quality is necessary to define the state of two phase mixture and derive an expression for specific volume of a two phase mixture in terms of quality.

B. Long Answer Questions (LAQs)

2081 End-Sem Q7 (7 Marks - Part):

Sketch the saturation curve of two phase mixture on P-v and T-v diagrams. Also label all important points, lines and regions.

2079 End-Sem Q7 (8 Marks - Part):

Fill in the blanks in the following table with the corresponding properties of water.

• *Table Data:*

- State 1: P = 300 kPa, T = 133°C, x = 82%, v = 0.5 m³/kg, h = 2346 kJ/kg. [Note: Some of these values are handwritten/filled as answers in the past paper.]
- State 2: P = 476 kPa, T = 150°C, x = 60%, v = 6.23, h = 1900 kJ/kg.
- State 3: P = 200 kPa, T = 200°C, x = blank, v = 0.0011 m³/kg, h = 855 kJ/kg.
- State 4: P = 1000 kPa, T = 179°C, x = 60%, v = 0.11 m³/kg, h = 2000 kJ/kg.

2079 End-Sem Q11 (7 Marks - Part):

A rigid vessel having a volume of 0.4 m^3 contains 2.0 kg of liquid water and water vapor mixture in equilibrium at a pressure of 250 kPa . Calculate: (i) The volume and mass of liquid, (ii) The volume and mass of vapor, (iii) Temperature, (iv) Enthalpy, (v) If it is heated until its pressure reached to 350 kPa , what will be its quality?

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Derivations of two-phase property relationships and extensive steam table calculations. You must be comfortable calculating v , h , u , and s using quality (x).
- **Direct Repetitions:**
 - Derivation of specific volume of two-phase mixture ($v = v_f + x v_{fg}$) is **REPEATED: 2079 Q3, 2080 Q3, and 2081 Mid-Term Q3.**
 - Rigid vessel steam property numericals (where volume is constant) are consistently featured across **2079 Q11, 2081 Q3, 2081 Mid-Term Q3, and 2082 Mid-Term Q3.**
- **Key High-Priority Topics:**
 - Definition of critical point, triple point, saturation state, quality (dryness fraction), and superheated region.
 - Derivation of $v = v_f + x v_{fg}$ and $h = h_f + x h_{fg}$.
 - Plotting Phase Diagrams (P - v , T - v , P - T) showing liquid, two-phase, and vapor regions.

Unit 4: First Law & Control Volume Analysis - [12% - 19%] of Total Marks

A. Short Answer Questions (SAOs)

2082 Mid-Term Q4 (6 Marks):

Define steady state work applications. Write down general energy equation for a turbine and compressor and reduce it for an adiabatic turbine and compressor.

2082 Mid-Term Q5 (6 Marks):

A gas turbine develops 60 MW of power output, mass flow rate of gas is found to be 80 kg/s . Properties of gas at inlet and exit of the turbine are as follows:

- *Inlet:* $P = 8 \text{ MPa}$, $T = 500^\circ\text{C}$, $V = 50 \text{ m/s}$, $z = 10 \text{ m}$.
- *Outlet:* $P = 0.1 \text{ MPa}$, $T = 50^\circ\text{C}$, $V = 150 \text{ m/s}$, $z = 5 \text{ m}$.

Determine the rate at which heat is lost from the turbine surface.

2081 End-Sem Q4 (8 Marks):

Air enters a compressor operating at steady state at 100 kPa, 300 K and leaves at 1000 kPa, 400 K, with a volumetric flow rate of 1.5 m³/min. The work consumed by the compressor is 250 kJ per kg of air. Neglecting the effects of potential and kinetic energy, determine the heat transfer rate, in kW. [Take $R = 287 \text{ J/kg K}$ and $c_p = 1005 \text{ J/kgK}$]

2081 Mid-Term Q4 (8 Marks):

A steam turbine develops 60 MW of power output. Mass flow rate of steam is found to be 80 kg/sec. Properties of steam at inlet and exit of the turbine as follows:

- *Inlet:* $P = 8 \text{ MPa}$, $T = 500^\circ\text{C}$, $V = 50 \text{ m/sec}$, $z = 10 \text{ m}$.
- *Exit:* $P = 0.4 \text{ MPa}$, $x = 80\%$, $V = 150 \text{ m/sec}$, $z = 5 \text{ m}$.

Determine the rate at which heat is lost from the turbine surface and determine the inlet and outlet areas.

2079 End-Sem Q5 (8 Marks):

Derive the general mass conservation and energy conservation equations for control volume.

B. Long Answer Questions (LAQs)

2081 End-Sem Q8 (7+8 Marks):

- **a.** Define steady state work applications. Write down the general energy equation for a turbine, compressor, heat exchanger and a nozzle.
- **b.** Steam enters a turbine operating at steady state condition with a mass flow rate of 1.2 kg/s. Properties of the steam at the inlet are $P_1 = 5 \text{ MPa}$, $T_1 = 450^\circ\text{C}$, $\bar{V}_1 = 10 \text{ m/s}$ and the exit are $P_2 = 100 \text{ kPa}$, $x_2 = 80\%$, $\bar{V}_2 = 50 \text{ m/s}$. If the power output of the turbine is 1200 kW, determine the rate of heat transfer from the turbine.

2080 End-Sem Q7 (8+7 Marks):

- **a.** State and explain conservation of energy for control volume.
- **b.** Steam enters through a turbine operating under steady state condition at a rate of 5 kg/s. The properties of steam at turbine inlet are $P_1 = 1 \text{ MPa}$ and $T_1 = 500^\circ\text{C}$ and at the turbine exit $P_2 = 50 \text{ kPa}$ and $x_2 = 0.9$. If the heat loss from the turbine surface occurs at a rate of 500 kW, determine the power output from turbine.

2080 End-Sem Q11 (7 Marks - Part):

Steam enters a nozzle operating at steady state with $P_1 = 10$ bar, $T_1 = 400^\circ\text{C}$ and velocity of 10m/s . The steam flows through the horizontal adiabatic nozzle. At the exit, $P_2 = 1.5$ bar and the velocity of 1068.13m/s . The mass flow rate is 2kg/s . Determine the exit area of the nozzle in m^2 . [Refer the table for properties of water]

2079 End-Sem Q10 (7 Marks - Part):

Air expands through an adiabatic turbine from 1000kPa and 1000K to 100kPa and 400K . The inlet velocity is 10m/s whereas exit velocity is 100m/s . The power output of the turbine is 3600kW . Determine the mass flow rate of air and the inlet and exit areas. [Take $R = 287\text{J/kgK}$ and $C_p = 1005\text{J/kgK}$]

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** This unit presents some of the most reliable scoring opportunities. It consistently features mathematical derivations of the Steady Flow Energy Equation (SFEE) followed by standard engineering applications: steam turbines, nozzles, or air compressors.
- **Direct Repetitions:**
 - "Derive general mass/energy conservation equations for control volume" was asked in **2079 Q5** and **2080 Q7(a)**.
 - SFEE reduction equations for turbines, compressors, and nozzles are **REPEATED** in **2081 Q8(a)** and **2082 Mid-Term Q4**.
 - The 60MW turbine numerical is highly recurring: **2081 Mid-Term Q4** is nearly identical to **2082 Mid-Term Q5** (differing only in outlet conditions: 2081 uses quality $x = 80\%$ whereas 2082 uses superheated temperature $T = 50^\circ\text{C}$).
- **Key High-Priority Topics:**
 - Derivation of SFEE starting from the Reynolds Transport Theorem or basic energy balances.
 - Applying the SFEE to a Turbine (Work output), Compressor (Work input), Nozzle (Kinetic energy gain), and Throttling Valve (Isenthalpic process).

Unit 5: Second Law of Thermodynamics - [12% - 14%] of Total Marks

A. Short Answer Questions (SAQs)

2081 End-Sem Q5 (8 Marks):

State Kelvin-Planck and Clausius statements of second law of thermodynamics. Also prove their equivalence.

2081 Mid-Term Q5 (8 Marks):

Explain the working principle of Carnot cycle with corresponding processes on P-v and T-s diagrams. Also explain its significance.

2080 End-Sem Q8 (8+7 Marks - Part):

Define heat engine, heat pump and refrigerator. Explain how first and second law is applied to determine performance of heat engine and refrigerator.

B. Long Answer Questions (LAQs)

2081 End-Sem Q9 (3+6 Marks - Part):

- **a.** Differentiate between reversible and irreversible process.
- **b.** A Carnot engine operates between two reservoirs at temperatures T_1 and T_h . The work output of the engine is 0.6 times the heat rejected. The difference in temperatures between the source and the sink is 200°C . Calculate the thermal efficiency, source temperature and the sink temperature. **[REPEATED: Similar values to 2080 Q8]**

2080 End-Sem Q8 (7 Marks - Part):

A Carnot engine operates between two reservoirs at temperature T_h and T_l . The work output of the engine is 0.6 times the heat rejected. The difference in temperature between the source and the sink is 200°C . Calculate the thermal efficiency, source temperature and the sink temperature.

2079 End-Sem Q8 (7 Marks - Part):

A heat pump having COP of 5 maintains a building at a temperature of 24°C by supplying heat at a rate of 72000kJ/h , when the surroundings is at 0°C . The heat pump runs 12 hours a day and electricity costs Rs 10/KWh. (i) Determine the actual and minimum theoretical cost per day. (ii) Compare the actual operating cost with the cost of direct electric resistance heating.

2079 End-Sem Q10 (4+4 Marks - Part):

State and explain Kelvin-Planck and Clausius statement of second law of thermodynamics. Also prove their equivalence.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Focuses on proving second-law equivalence (Kelvin-Planck vs. Clausius) and evaluating thermal cycles using Carnot limits.
 - **Direct Repetitions:**
 - "State Kelvin-Planck and Clausius statements and prove their equivalence" is **REPEATED in 2079 Q10 and 2081 Q5.**
 - The Carnot engine numerical (work output is 0.6 times heat rejected, temperature difference is 200°C) is **REPEATED with identical values in 2080 Q8 and 2081 Q9.**
 - **Key High-Priority Topics:**
 - Mathematical proof of equivalence between Kelvin-Planck and Clausius statements (by creating a violator device).
 - Definitions of Heat Engine thermal efficiency (η), and COP for Refrigerator (COP_R) and Heat Pump (COP_{HP}).
 - Carnot principles and Carnot cycle processes on P-v and T-s coordinates.
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Unit 6: Entropy - [5% - 7%] of Total Marks

A. Short Answer Questions (SAQs)

2082 Mid-Term Q6 (6 Marks):

Derive the entropy relations for an ideal gas and incompressible fluid or solid.

B. Long Answer Questions (LAQs)

2081 End-Sem Q9 (6 Marks - Part):

Define entropy and derive the entropy relations for an ideal gas.

2080 End-Sem Q11 (8 Marks - Part):

Derive expressions for change in entropy for an ideal gas and incompressible substance.

2079 End-Sem Q7 (7 Marks - Part):

Define entropy and isentropic process. Derive the mathematical expression for entropy relation for an ideal gas in terms of pressure and temperature and temperature and volume.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Purely analytical and mathematical. Every single paper asks for the exact same derivation: the entropy change for an ideal gas.
 - **Direct Repetitions:**
 - Derivation of entropy change for an ideal gas is **REPEATED: 2079 Q7, 2080 Q11, 2081 Q9, and 2082 Mid-Term Q6.**
 - **Key High-Priority Topics:**
 - Clausius Inequality ($\oint \delta Q/T \leq 0$) as the foundation of entropy.
 - Ideal gas entropy change derivations using $T ds$ relations:
 1. $s_2 - s_1 = C_v \ln(T_2/T_1) + R \ln(v_2/v_1)$
 2. $s_2 - s_1 = C_p \ln(T_2/T_1) - R \ln(P_2/P_1)$
 - Isentropic process definition ($ds = 0$ for reversible adiabatic processes).
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Unit 7: Thermodynamic Cycles - [19% - 25%] of Total Marks

A. Short Answer Questions (SAQs)

2081 End-Sem Q6 (8 Marks):

Explain the working principle of an ideal Rankine cycle. Sketch the cycle on P-v and T-s diagrams and derive an expression for its efficiency.

2079 End-Sem Q4 (8 Marks):

A power-plant operating on an ideal Brayton cycle delivers a power output of 80MW. The minimum and maximum temperatures during cycle are 300K and 1500K respectively. The pressure at the inlet and exit are 100kPa and 1400kPa respectively. (i) Determine the thermal efficiency of the cycle, (ii) Determine the power output from the turbine, and (iii) What fraction of the turbine power output is required to drive the compressor? [Take $C_p = 1005 \text{ J/kg K}$, $\gamma = 1.4$]

B. Long Answer Questions (LAQs)

2081 End-Sem Q10 (8+7 Marks):

- a. Sketch the Diesel cycle on P-v and T-s diagrams and derive an expression for its efficiency in terms of compression ratio and cut-off ratio.

- **b.** An air standard Otto cycle has a compression ratio of 8 and before compression stroke has air at 100kPa and 300K. The peak pressure during cycle is 6000kPa. Determine: (i) the peak temperature in the cycle, (ii) the temperature at the end of expansion stroke, (iii) the cycle efficiency.

2080 End-Sem Q6 (8 Marks):

The compression ratio of an air standard Otto cycle is 8. At the beginning of the compression process, the pressure and temperature of air are 100kPa and 22°C respectively. The heat added per kg of air during the cycle is 2100kJ/kg. Determine: (i) The pressure and temperature at the end of each process of the cycle, (ii) The thermal efficiency, (iii) The mean effective pressure.

[Take $R = 287 \text{ J/kgK}$ and $c_v = 718 \text{ J/kg K}$]

2080 End-Sem Q9 (7+8 Marks):

- **a.** Draw the layout diagram, P-v and T-s diagram of air standard diesel cycle and derive an expression for its efficiency in terms of compression ratio and cut-off ratio.
- **b.** An ideal Brayton cycle has pressure ratio of 10. The temperature of air at the compressor and turbine inlets are 300K and 1200K respectively. Determine the thermal efficiency and mass flow rate of air to produce net power out of 80MW. [Take $C_p = 1005 \text{ J/kgK}$, $\gamma = 1.4$]

2079 End-Sem Q8 (8 Marks - Part):

Define power cycle and refrigeration cycle. Sketch the Rankine cycle on P-v and T-s diagrams and derive the expression for efficiency.

2079 End-Sem Q9 (7 Marks - Part):

The compression ratio of an air standard Otto cycle is 8. At the beginning of the compression process, the pressure and temperature of air are 100kPa and 20°C respectively. The heat added per kg of air during the cycle is 2000kJ/kg. Determine: (a) the pressure and temperature at the end of each process of the cycle, (b) the thermal efficiency and (c) the mean effective pressure.

[Take $\gamma = 1.4$, $C_v = 718 \text{ J/kg K}$]

2079 End-Sem Q11 (8 Marks - Part):

Explain the working principle of an ideal Diesel cycle. Sketch the cycle on P-v and T-s diagrams and derive expressions for its efficiency in terms of compression ratio and cut-off ratio.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Heavy on cycle efficiency derivations and complete thermodynamic process loop evaluations. You must know how to solve Otto, Diesel, Brayton, and Rankine cycles.
 - **Direct Repetitions:**
 - Derivation of Diesel cycle efficiency in terms of compression ratio (r) and cut-off ratio (r_{c}) is **REPEATED: 2079 Q11, 2080 Q9(a), and 2081 Q10(a).**
 - Derivation of Rankine cycle efficiency is **REPEATED: 2079 Q8 and 2081 Q6.**
 - The Otto cycle numerical with a compression ratio of 8 is **REPEATED: 2079 Q9, 2080 Q6, and 2081 Q10(b).**
 - **Key High-Priority Topics:**
 - Layout diagrams and P-v/T-s curve sketching for Otto, Diesel, Brayton, and Rankine cycles.
 - Solving for temperature and pressure values at every corner state of air standard cycles (Otto and Diesel).
 - Deriving thermal efficiency equations for all four core cycles.
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Unit 8: Heat Transfer - [12% - 13%] of Total Marks

A. Short Answer Questions (SAOs)

2079 End-Sem Q6 (8 Marks):

A steel pipe having an outside diameter of 2 cm is to be covered with two layers of insulation, each having a thickness of 1 cm. The average conductivity of one material is 5 times that of the other. Assuming that the inner and outer surface temperatures of the composite insulation are fixed, calculate by what percentage of heat transfer will be reduced when the better insulating material is next to the pipe than it is away from the pipe.

B. Long Answer Questions (LAQs)

2081 End-Sem Q11 (8+7 Marks):

- **a.** Define heat transfer and different modes of heat transfer. Derive an expression for steady state heat transfer through a composite plane wall consisting of three layers of different materials.

- **b.** A steel pipe with inner diameter and outer diameter as 80 mm and 120 mm is covered with two layers of insulation, 25mm and 40mm thick. The thermal conductivities of insulating materials are 0.2 W/mK and 0.1 W/mK respectively while that of steel is 50W/mK. The inner surface of the pipe is 200°C while a surface temperature of insulation is 40°C. Determine the heat loss from the unit length of the pipe and layer contact temperatures.

2080 End-Sem Q10 (7+8 Marks):

- **a.** Derive an expression with appropriate diagram for conduction heat transfer through a composite cylinder tube consisting of three layers of different materials.
- **b.** A hollow cylinder with inner and outer diameter of 8 cm and 12 cm respectively has an inner surface temperature of 200°C and outer surface temperature of 50°C. If the thermal conductivity of the cylinder material is 60W/mK, determine the heat transfer from the unit length of pipe. Also determine the temperature at the surface at a radial distance of 5cm from the axis of the cylinder.

2079 End-Sem Q9 (8 Marks - Part):

Define the three modes of heat transfer. Derive an overall heat transfer coefficient for a hollow cylinder subjected to convection medium on both sides.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Balanced between solid conduction derivations (plane composite walls or hollow composite cylinders) and high-value numericals modeling radial insulation heat losses.
- **Direct Repetitions:**
 - Derivation of conduction heat transfer through composite cylinders/convective boundaries is **REPEATED: 2079 Q9 and 2080 Q10(a).**
 - Thermal resistance circuit evaluation for steel pipes covered in insulation layers is highly recurring: **2079 Q6, 2080 Q10(b), and 2081 Q11(b).**
- **Key High-Priority Topics:**
 - 1D steady-state heat conduction derivations: Fourier's law applied to plane composite walls ($q = \Delta T / \Sigma R_{th}$) and composite radial pipe geometries ($q = 2\pi L(T_i - T_o) / \Sigma [\ln(r_{j+1}/r_j)/k_j]$).
 - Evaluating interface/contact temperatures at the boundary surfaces between composite layers.
 - Distinguishing the physical modes: Conduction, Convection.

SUBJECT PRIORITY PLANNER (THERMODYNAMICS)

Rank	Unit No.	Unit Name	Exam Priority Class	Historical Mark Percentage	Core Focus Areas to Pass/Score
1	Unit 7	Thermodynamic Cycles	Highest	19% - 25%	Otto/Diesel process paths and efficiency derivations, Brayton/Rankine efficiency proofs, thermal efficiency calculations.
2	Unit 4	First Law & CV Analysis	Highest	12% - 19%	General energy equation derivations for open control volumes, SFEE reductions for turbines/nozzles, turbine energy balance calculations.
3	Unit 3	Properties of Pure Substance	High	12% - 19%	Steam table calculations, derivation of specific volume relationship using dryness fraction (x), two-phase state property determinations.
4	Unit 5	Second Law of Thermodynamics	High	12% - 14%	Refrigerator/Heat pump performance analyses, equivalent statements equivalence proof (KP vs. Clausius), Carnot engine thermal efficiency calculations.
5	Unit 2	Energy and Energy Transfer	High	8% - 13%	Path vs. point functions, spring-piston displacement work calculations, work transfer derivations for isobaric/polytropic paths.
6	Unit 8	Heat Transfer	Medium-High	12% - 13%	Radial 1D conduction proofs through composite insulation tubes, overall heat transfer coefficient calculations, convective boundary equations.
7	Unit 1	Introduction and Basic Concepts	Medium	5% - 7%	Macroscopic vs. microscopic views, Zeroth law conceptual significance, intensive vs. extensive property evaluations.
8	Unit 6	Entropy	Low	5% - 7%	Ideal gas entropy change equations derivation (T ds relations), isentropic state definition.